

SMITHIAN FOLD THEORY - COMPUTATIONAL PROOF PAPER 002

From Attention to an Exact Conversational Architecture

Unison Fold AI: a V3-native translation of a GPT-2-class causal conversational
transformer

Architecture proof, clean-room implementation, zero-copy teacher distillation and an
open conversational gate

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OPEN-ACCESS PRELIMINARY VERSION - RESULTS IN PROGRESS

Abstract

Conversational language modelling tests whether an architecture can preserve finite symbol identity, causal order, selective attention, bounded memory, learning and generation while remaining inspectable at its mathematical boundary. This paper reports the preliminary Smithian Fold Theory (SFT) V3 Unison Fold AI computational proof programme. Its one-to-one requirement is narrow and explicit: the functional operation roles of a GPT-2-class attention-based causal transformer are translated into the registered SFT mathematical domain. It is not a copy of Unison AI V2, GPT-2 code, GPT-2 weights, GPT-2 dimensions, or GPT-2 numerical functions.

The registered operation inventory contains 40 rows. Thirty-eight cover the reference tokenizer, parameter, representation, transformer, training and generation path; two are downstream conversational and Discord application operations. Every row names an SFT form, interface, dependency route and required falsification check. Existing project records classify the 40-row implementation as passing at its declared finite operation boundary. The architecture retains a fresh reversible byte-BPE tokeniser. The architecture also retains bounded causal context, token and position carriers, pre-layer normalisation, joint query/key/value projection, masked multi-head attention, residual junctions, a nonlinear feed-forward path, a tied language-model head, causal learning, cached autoregression and finite generation. Conventional exponential softmax and tanh GELU are not imported as proof premises. Their function roles are translated respectively by exact finite normalisation and an oriented rational gate, and numerical equality with GPT-2 is not claimed.

The translation grammar crosses eight binary mathematical choices. All $2^8 = 256$ candidates were generated, all were decided, 255 were eliminated, and the unique preserving survivor is **UNISON-TRANS-001**. The current dependency inventory contains 27 registered SFT claims spanning exact mathematics, finite probability, computation, information, learning, attention and memory. The application imports those relations downstream; it does not alter them, admit a new SFT theorem or law, or confer scientific authority on an engineering artefact.

An identified exact native artefact demonstrates the declared proof-and-trace mechanism at one finite engineering scale: vocabulary 769, context 128, eight layers, eight heads, head width 16, embedding width 128, feed-forward width 512 and 1,701,248 trainable exact scalars. Its floating proposal was accepted only after quantisation into the exact carrier reduced the registered exact objective from 64676464973/65536 to 5364696355/262144. A separate standard-library verifier reconstructed the forward path without importing the runtime and reported a pass. Neither result establishes conversational readiness.

The empirical record is adverse and remains visible. Eleven successive native candidates, including the exact artefact above, were rejected for hand-off after unseen development conversations exposed fragmentation, malformed words, weak relevance, repetition, truncation or failure of direct and multi-turn fact grounding. A fresh student is therefore being trained from an independently initialised native carrier using the official frozen GPT-2 124M model as an offline teacher. The teacher has 124,439,808 parameters; exactly zero are copied. Its full next-symbol distributions are contracted into 257 tokeniser-independent byte/EOS groups on a 2^{24} mass grid. The current receipt binds 32,763 selected signals from 8,192 assistant-turn examples. Independent unique-source_row_id aggregation over the retained signal corpus yields 6,121 distinct dialogue rows; that count is recorded in the continuation checkpoint rather than in the receipt itself. The teacher is neither a runtime dependency nor a fallback, and its use is learning evidence rather than derivational authority.

At this paper's frozen evidence boundary, the fresh distillation run is still in proposal training. The snapshot at 2026-07-31T17:23:43.846494+00:00 retains checkpoint 3,200 of 16,000; later live checkpoints belong to a future successor version. The active run has no final native bundle identity, no exact acceptance decision, no independent verification, no passing unseen conversation record, no completed GPT-2 comparison and no manual Discord result. The registered Discord channel already receives sanitised training/debugging progress and automatically refreshed ETA events, but the connection is output-only while the candidate remains unaccepted. This paper will be amended by explicit publication update after later results exist. No GPT-2 parity, conversational upgrade, generalised conversational validation, public deployment, scientific admission or completed theory-of-everything claim follows from the present preliminary evidence.

Detailed preliminary-results synopsis

The programme begins from a separation that earlier Unison work did not hold reliably: architecture, learning evidence, conversational evidence and transport evidence are different objects. A Discord reply can show that a socket works without showing that a transformer organ exists. An exact forward trace can show that an attention mechanism executes without showing that its learned distribution supports a useful conversation. A frozen teacher can provide behavioural supervision without becoming the student's architecture, parameters or runtime. The V3 programme therefore binds each statement to the smallest evidence class that warrants it.

The architecture reference is the frozen published and archived GPT-2 functional path. The reference may supply operation names, order, tensor interfaces and causal behaviour. It may not supply SFT laws, survivor selection, source code for the V3 implementation, vocabulary, merge ranks, pretrained parameters, answers or runtime responses. Unison AI V2 is even narrower: it is a read-only observational comparison object used to retain earlier failure modes. V2 is not the template being reproduced.

The SFT translation places every proof-facing scalar in a carrier of structural absence or an oriented positive rational part. Counts, labels, positions, words, relations and generated finite supports remain explicit. Future attention positions are structurally excluded rather than hidden under a large negative sentinel. Selected and unselected attention and generation alternatives remain reconstructable on the registered exact proof path. The exact attention scale requires an integral square root of head width. Layer-normalisation square roots are held by rational enclosure.

The candidate census makes these commitments testable. It combines the allowed and prohibited choices for scalar carrier, attention normalisation, causal masking, head scaling, feed-forward gating, layer-normalisation measure, weighted-generation cause and key/value memory. The sole survivor selects the permitted choice on every axis. Uniqueness is therefore uniqueness within this complete registered eight-axis grammar; it is not a claim that no other conceivable conversational architecture can exist.

The old widest exact artefact is important because it separates proof of mechanism from evidence of utility. Its exact model, tokenizer and bundle identities are preserved. Its learning trace retains every exact acceptance example and before/after objective. Its independent verifier reconstructs eight layers, eight heads and 1,538 logits without importing the runtime module. Yet its unseen replies remained malformed and irrelevant, and both the direct and multi-turn green-colour tests failed. The artefact is consequently verified as an implementation and rejected as a conversational hand-off candidate.

The current rebuild addresses quality through learning rather than puppetry. The native student begins from a fresh independently caused parameter carrier. DailyDialog contributes 10,862 development conversations from its official 11,118-row training partition; a disjoint 256-row development probe remains frozen, and the validation and test archives remain unopened. Eight project-authored mechanism dialogues exercise exact construction and acceptance. No corpus row is used as a response lookup, template or fallback.

The frozen GPT-2 teacher supplies offline next-byte-distribution evidence. For each selected response-prefix position, its complete vocabulary distribution is contracted into 256 possible first bytes plus EOS. Largest-remainder quantisation preserves a complete finite mass of 16,777,216 parts. GPT-2 token identities, merge ranks, logits and parameters do not enter the native bundle. After a candidate is accepted, its persistent bundle can be loaded in later sessions without GPT-2, Transformers or the teacher cache. A later, larger open teacher may continue that accepted lineage only under a new registered receipt that preserves the parent and still copies zero parameters.

The route to manual testing is therefore finite but unfinished. The active build must become a fixed native artefact, pass the exact objective and independent verifier, pass the frozen seven-case offline conversation gate and cold review, complete the registered GPT-2 comparison or retain an explicit unavailable result, and pass the final seal/provenance/secret audit. Only then may the accepted model be launched as the conversational process in Discord for Maria Smith's first private observation.

Headline findings

Result	Exact current record	Scientific status
Operation-role inventory	40 rows	Complete at the registered inventory boundary
Reference/application split	38 reference-path rows; 2 downstream conversation rows	Explicit; conversation rows are not attributed to GPT-2
Translation census	8 binary axes; 256 candidates; 255 rejected; 1 survivor	Exact closure within the registered grammar
Direct SFT dependency inventory	27 registered claims	Existing project inventory at the frozen paper boundary
Native tokenizer	Fresh reversible byte-BPE	No GPT-2 vocabulary or merge ranks copied

Result	Exact current record	Scientific status
Widest verified exact artefact	8 layers; 8 heads; width 128; 1,701,248 trainable scalars	Mechanism verified; conversational hand-off rejected
Exact objective	64676464973/65536 before; 5364696355/262144 after	Exact candidate acceptance at the registered development boundary
Independent reconstruction	8 layers; 8 heads; 1,538 logits; runtime not imported	Pass for the identified old artefact
Adverse conversation record	11 native candidates rejected	Conversational readiness not established
DailyDialog custody	11,118 train rows; 10,862 development; 256 frozen probe	Validation/test archives unopened; local non-commercial use
Teacher custody	GPT-2 124M; 124,439,808 parameters	Offline learning instrument only
Soft teacher support	32,763 receipt-bound signals; 8,192 receipt-bound assistant-turn examples; 6,121 distinct dialogue rows derived from the retained signal corpus; 257 groups; denominator 16,777,216	Receipt plus independently derived corpus aggregation
Teacher parameter copy	0	Native student remains independently parameterised
Runtime teacher dependency	false	Accepted native bundle is designed to persist without teacher access
Workspace verification	56 tests passed; 0 failed/error	Existing implementation and protocol checks
Discord channel policy	1523685773998555227 only	Offline routing tests pass; progress relay connected output-only
Fresh distilled candidate	Proposal training in progress	No result claimed before its final receipts and gates
Frozen unseen conversation	No passing candidate	Open
GPT-2 comparator	Protocol and environment registered; execution pending	Open
Manual Discord test	Not run	Open
SFT admission	No new admission requested or awarded	Application remains downstream
Remote publication	Authorised for version 0.1.0-rc1	Zenodo DOI 10.5281/zenodo.21726397 and GitHub main publication

Current status

Formally specified. The complete 40-row operation-role inventory, 27-claim dependency surface and 256-form translation census are present. The unique survivor is exact within the registered grammar.

Implementation and checks. Existing project records classify the 40-row implementation as passing at its declared finite operation boundary.

Computationally demonstrated. An identified native artefact executes the declared exact causal-transformer proof path and passes the registered exact objective plus an implementation-distinct forward verifier. This is an artefact-bounded mechanism result, not a claim about all possible scales.

Empirically adverse so far. Eleven candidates failed the frozen development-visible conversation boundary. No candidate has passed the direct and multi-turn grounding requirement.

Currently running. A newly initialised student is undergoing offline GPT-2 teacher distillation. Mutable losses, checkpoint projections and ETA events belong to the operational terminal stream. Version 0.1.0-rc1 freezes checkpoint 3,200 of 16,000 at 2026-07-31T17:23:43.846494+00:00; later events do not revise this snapshot. They are not a paper result.

Discord status. The sole registered channel receives sanitised progress events. Conversational input remains disabled for the unaccepted candidate. No manual test has occurred.

Scientific and publication boundary. This is an SFT V3 frontier application. It establishes no new admitted theorem or law, no GPT-2 parity, no generalised conversation result, no consciousness claim and no public deployment. Maria Smith retains publication authority. She authorised this version's GitHub main push and Zenodo publication on 31 July 2026.

1. Public scientific mission and authorship boundary

Ernos Labs is an open-science movement, verification platform and public tree of knowledge founded by Maria Smith. Its purpose is to make scientific authority narrow, inspectable and revocable. Derivations, candidate spaces, eliminations, source custody, executable traces, controls and adverse results should remain available to criticism. Credentials may route review; they do not replace evidence.

Maria Smith developed Smithian Fold Theory outside the conventional academic and institutional route. That history is not evidence for this application. It motivates the institutional boundary: neither prestige nor automation may admit a scientific claim. Automated agents may provide code, editorial work and checks, but they receive no scientific authorship, ownership, admission authority or publication authority.

Maria Smith is the sole scientific author and publication authority for this paper. She explicitly authorised the commit and direct GitHub `main` push of this paper package and its publication as a standalone Zenodo preliminary record. That authority applies only to version 0.1.0-rc1 and does not authorise future versions or public deployment of the conversational bot.

2. Scope and ownership boundary

The application scope is a text-only, decoder-only conversational model served through one private Discord channel. Its first intended user-facing boundary is basic casual back-and-forth with direct and multi-turn context retention. Voice, images, scraping, autonomous tools, unsolicited messages, public deployment and public-channel operation are excluded.

The one-to-one statement applies only to functional operation roles in the registered GPT-2-class attention transformer path. It does not require:

- the V2 architecture or behaviour;
- GPT-2 source code;
- GPT-2 parameter values;
- GPT-2 vocabulary or byte-pair merge ranks;
- GPT-2 tensor dimensions or parameter count;
- exponential-softmax or tanh-GELU numerical equality;
- output-by-output identity with GPT-2; or
- a claim that GPT-2 itself is an SFT derivation.

Mathematics owns exact carriers, maps, contractions, finite distributions and enclosures. Computation and Information own words, encodings, causal order, finite memory, transforms and state transitions. Learning owns target, objective, update and held-out custody. The SFT Attention and Memory claims own retained selection alternatives, capacity and renewal. Unison Fold AI composes these admitted relations as a downstream application. Discord owns only the transport boundary.

3. Terminology and evidence constitution

A **theorem** is an exact formally admitted result. A **law** is an admitted dependency relation. A **claim** is a registered statement with formal and empirical status. An **application computational proof** is an exact, reconstructable implementation consequence of registered dependencies at a declared finite boundary; it is not automatically a new model-admitted claim.

A **derivation** is generated from declared carriers and relations. A **translation** preserves a registered function role while replacing a mathematical form that lies outside the SFT domain. A **correspondence** does not imply numerical identity. A **prediction** is a prospectively sealed consequence evaluated only after its target becomes accessible. A **development result** may guide engineering but cannot be backdated into a blind validation.

Verification means that code, an artefact, trace or receipt satisfies its registered mechanical checks. **Validation** is reserved for the applicable empirical protocol. **Conversational readiness** requires relevant, non-fragmented exchanges and registered context grounding; a non-empty reply is insufficient. **GPT-2 parity** or **upgrade** would require the separate frozen comparison rule and is not established here.

Formal status, computational status, empirical status and **publication status** remain distinct. Test success cannot substitute for conversation evidence, and conversation evidence cannot alter a mathematical dependency. Every use of *complete* in this paper names the object completed.

4. Scientific question and one-to-one reference boundary

The scientific application question is:

<i>Can the complete functional path of a GPT-2-class causal transformer be</i>
<i>reconstructed in the declared finite SFT mathematical domain, trained</i>
<i>without copied pretrained parameters, persisted as a native artefact and</i>
<i>tested as a conversational Discord system?</i>

The frozen GPT-2 paper and archived implementation establish the reference-side inventory. They may identify reversible byte encoding, ranked byte-pair merging, token and position embeddings, pre-normalised transformer blocks, joint Q/K/V projections, scaled causal multi-head attention, residual connections, the feed-forward nonlinearity, a tied language-model head, causal next-token learning and cached autoregressive generation.

Reference custody stops there. The SFT side must state its own carriers, relations and constraints. The architecture is one-to-one by registered function role, not by floating formula or stored number. Distillation is a later learning process and cannot repair a missing operation row.

5. Mathematical constitution

5.1 Exact scalar and tensor carrier

Every proof-facing scalar is either structural absence or an orientation paired with a positive rational magnitude:

OrientedPart = absent | (positive, q) | (negative, q), where q is a positive fraction.

Host-language integer zero may implement absence but is not serialised as an SFT number. Tensor shapes, indices, positions, layer identities, head identities and vocabulary identities are generated finite carriers. No floating equality, completed infinity or hidden irrational value is accepted as proof evidence.

5.2 Exact finite attention normalisation

For the complete causally available compatibility support, let m be the retained minimum and let d_i be the exact oriented distance of score i above m . The translation assigns:

$$w_i = (d_i + \text{One}) / \sum_j (d_j + \text{One}).$$

Every available item receives a positive part, ties remain ties, ordering is monotone and the complete support composes exactly to One. Future positions are absent from the available support but remain named in the exact proof-trace exclusion ledger. This `FINITE_NORMALIZE` operation serves the role of attention normalisation; it is not numerically equal to exponential softmax.

5.3 Oriented rational feed-forward gate

The pointwise nonlinear organ is retained by `ORIENTED_GATE`. For a positive input of magnitude q , the gate part is $(\text{One} + q) / (\text{Two} + q)$. For a negative input it is $\text{One} / (\text{Two} + q)$. Structural absence remains absent. Multiplication by the input preserves orientation and supplies a rational nonlinearity. It is a function-role translation, not a numerical claim about tanh GELU.

5.4 Normalisation, scaling and causal exclusion

Layer normalisation retains exact mean centring, a rational square-root enclosure and the selected rational representative before applying its affine scale and bias. Attention divides by the registered integral square root of `head_width`; a non-square head width is rejected at this boundary. Causal masking names every future candidate and records why it is excluded rather than assigning a negative numeric sentinel.

5.5 Ordered model path

For one finite token word, the sole model path is:

TOK → CTX → token/position REP → [BLK × layer_count] → final LN → tied HEAD → causal selection → append.

Each block is:

LN → QKV → split heads → causal attention → merge → projection → residual → LN → expansion → oriented gate → projection → residual.

Existing readiness audit v1 records serving and exact learning observations as invoking the same active native transformer path. This paper reports that recorded boundary without modifying it. A response table, scripted answer, teacher reply or fallback would be a falsifying substitution.

6. Dependency structure

The declared direct dependency inventory contains 27 registered SFT claims:

Domain	Claim identities	Application role
Formal computation	SFT-COMP-FORM-LANGUAGE-GRAMMAR-001; SFT-COMP-ALG-STRINGS-SEQUENCES-001; SFT-COMP-FORM-COMPOSITION-001; SFT-COMP-FORM-STATE-TRANSITION-001	Generated words, ordered positions, typed composition and successor traces
Causality and memory	SFT-COMP-DIST-CAUSALITY-001; SFT-COMP-ALGX-STREAMING-MEMORY-027; SFT-INFO-SOURCE-FINITE-MEMORY-010; SFT-CONSC-MEMORY-PERSISTENCE-001	Predecessor-only support, bounded context and renewed cache identity
Information and encoding	SFT-INFO-COMP-TRANSFORM-005; SFT-INFO-ENCODING-DECODING-001; SFT-INFO-SOURCE-SEQUENCE-ORDER-002; SFT-INFO-SYMBOL-DISTINCTION-001	Reversible transform, source order and symbol identity
Exact mathematics	SFT-MATH-EXACT-ARITHMETIC-001; SFT-MATH-NUM-EXACT-REPRESENTATION-ROUNDING-001	Rational operations and enclosure custody
Linear structure	SFT-MATH-LINEAR-VECTOR-COORDINATE-CARRIERS-001; SFT-MATH-LINEAR-MAP-COMPOSITION-002; SFT-MATH-LINEAR-INNER-PRODUCT-METRIC-008; SFT-MATH-LINEAR-TENSOR-CONTRACTION-012	Embeddings, affine maps, Q/K/V rows, compatibility and contractions
Finite probability	SFT-MATH-PROB-FINITE-DISTRIBUTION-006	Exact complete distributions and declared-cause cumulative selection
Attention	SFT-CONSC-ATTENTION-001; SFT-CONSC-ATTENTIONAL-CAPACITY-001	Selected, unselected, missed and competing support
Learning	SFT-COMP-LEARN-CLASSICAL-LEARNING-001; SFT-COMP-LEARN-CLASSIFICATION-PREDICTION-001; SFT-COMP-LEARN-LEARNING-OPTIMIZATION-001; SFT-COMP-LEARN-REPRESENTATION-001; SFT-COMP-LEARNX-HELD-OUT-CUSTODY-004; SFT-COMP-LEARNX-LEARNED-VERIFICATION-024	Examples, target custody, objective, representations, holdouts and independent verification

The inventory records 27 claims at this frozen boundary. This paper does not amend it or state that these claims exhaust SFT or conversational science.

7. Registered source surface

Source	Frozen identity	Permitted evidential role	Prohibited role
Current SFT V3 claim packages	Claim IDs in Section 6	Dependency relations and exact boundaries	Outcome-seeking alteration by this application
GPT-2 paper	SHA-256 d9d852e2894556e73f53cb22b7c605a9643d6f0b19bf604b429ed6192fa24f4e	Published functional inventory and scale context	SFT premise or native parameter source

Source	Frozen identity	Permitted evidential role	Prohibited role
OpenAI GPT-2 repository	revision 9b63575ef42771a015060c964af2c3da4cf7c8ab	Frozen operation order and interfaces	Copied V3 code
Frozen GPT-2 124M cache	weight SHA-256 248dfc3911869ec493c76e65bf2fcf7f615828b0254c12b473182f0f81d3a707	Offline teacher; later comparator under separate custody	Student initialisation, tensor copy, runtime or fallback
DailyDialog train archive	SHA-256 6191257f8250da0e045bfd1d017645e15b2358df975ec4c5c431a5a1f99662a4	Local non-commercial proposal-learning rows	Retrieval, public redistribution outside licence, hidden evaluation
Unison AI V2 records	Hash-bound read-only registry	Historical questions and adverse controls	Architecture authority, code, weights, answers or proof imports

All source prestige is irrelevant to formal admission. Each source is used only for the stated role. A successful transport does not prove a claim, and a failed transport cannot be silently relabelled as absence.

8. Translation candidate census

The translation census crosses eight independently named binary axes:

Axis	Preserving choice	Rejected choice
Scalar carrier	oriented positive part or structural absence	signed real number
Attention normalisation	complete exact finite parts	floating exponential softmax
Causal mask	retained structural exclusion	negative numeric sentinel
Head scale	integral-root exact division	imported irrational square root
Feed-forward gate	oriented rational gate	tanh-GELU real value
Layer-normalisation measure	rational square-root enclosure	unrecorded real square root
Weighted-generation cause	declared exact cause part	hidden ontic randomness
Key/value memory	renewed identity-retaining cache	opaque cache

The complete product has cardinality:

$2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256.$

Candidate UNISON-TRANS-001 chooses the preserving form on all axes. Every other candidate violates at least one declared constraint. The census records 256 decisions, 255 eliminations and one survivor. Its canonical census identity is sha256:019797d71296fcc4e8dfffa878e3c9d84ab8f70eebaa19dd187b28abd89b80346.

This result proves uniqueness only within the preregistered eight-axis grammar. It neither proves universal architectural uniqueness nor selects a model because of conversational scores.

9. Complete 40-row operation-role inventory

The following table is the conceptual view of the complete registered map. The source specification preserves the full dependencies and falsification checks for each row. The table records implementation roles, not a claim that each row already has its own distinct direct test.

ID	Reference or application role	V3 SFT form
TOK-01	Reversible byte carrier	Complete 256-byte internal carrier; valid UTF-8 source-sequence round trip
TOK-02	Unicode-aware segmentation	Ordered source spans retaining boundaries
TOK-03	Ranked byte-pair merging	Complete adjacent-pair census and replay trace

ID	Reference or application role	V3 SFT form
TOK-04	Token/id encode and decode	768 text/byte-BPE identities retain exact byte payloads; structural EOS is separate and has no text inverse
CTX-01	Bounded causal context	Active predecessor word with bounded retained conversation state
PAR-01	Learned parameter carrier	Named finite tensors of exact oriented parts
PAR-02	Residual-projection initialisation scale	Generated exact cause and admitted integral-root scaling
REP-01	Token embedding	Token identity selects a registered row
REP-02	Position embedding	Successor position selects a registered row
REP-03	Token/position addition	Exact coordinate-wise oriented junction
LN-01	Pre-block and final layer normalisation	Mean centring and rational square-root enclosure
ATT-01	Joint Q/K/V projection	Three named affine-map slices
ATT-02	Split and transpose heads	Bijective identity-retaining reshape
ATT-03	Concatenate cached past K/V	Ordered recurrence junction with source identity
ATT-04	Query/key compatibility	Exact complete dot-product contraction
ATT-05	Head-width square-root scaling	Exact division by registered integral root
ATT-06	Lower-triangular causal mask	Structural future exclusion ledger
ATT-07	Compatibility normalisation	Exact <code>FINITE_NORMALIZE</code> distribution
ATT-08	Attention-weighted value sum	Complete exact weighted contraction
ATT-09	Merge heads	Inverse of the registered split
ATT-10	Attention output projection	Named exact affine map
RES-01	Residual connection	Identity-preserving coordinate junction
MLP-01	Feed-forward expansion	Exact affine map to registered width
MLP-02	Pointwise GELU role	Exact rational <code>ORIENTED_GATE</code> correspondence
MLP-03	Feed-forward projection	Exact affine map back to residual width
BLK-01	Pre-normalised transformer block	Ordered $LN \rightarrow ATT \rightarrow RES \rightarrow LN \rightarrow MLP \rightarrow RES$ composition
STK-01	Repeated blocks	Finite successor-indexed composition
HEAD-01	Final normalisation	LN-01 after the last block
HEAD-02	Tied token projection	Contraction with the token-embedding identity
LOSS-01	Causal shift	Position i predicts successor $i+1$ only
LOSS-02	Next-token objective	Exact target-rank/support objective
LRN-01	Parameter update	Registered exact update over development rows
GEN-01	Prefix/cache autoregression	Append and renew bounded K/V relation
GEN-02	Temperature	Exact positive rational score scaling
GEN-03	Top-k restriction	Exact ordered support with exclusions retained
GEN-04	Nucleus restriction	Exact cumulative finite-mass support
GEN-05	Multinomial selection	Exact cause part against a cumulative finite distribution
GEN-06	Append and terminate	EOS or positive finite length bound
CONV-01	Conversational serialisation	Reversible role, speaker, turn and text word
CONV-02	Discord state	Single-channel ordered ledger and terminal custody

GEN-02 through GEN-05 are generation policies in the registered reference path and are not asserted to be transformer organs. CONV-01 and CONV-02 are downstream application operations and are not attributed to GPT-2. This distinction prevents the phrase “40 translations” from becoming a claim that GPT-2 contains Discord.

Existing project records classify all 40 rows as implemented and independently reconstructed for identified artefacts. This paper reports that status without changing the verification specification.

10. Native exact transformer implementation

The current implementation supplies a fresh native text/byte-BPE tokenizer, exact transformer carrier, fixed-grid serving route, exact learning records, conversation-state ledger and thin Discord adapter. The active model path has no production fallback generator.

The widest independently verified artefact has:

Component	Registered value
Vocabulary size	769: 768 text/byte-BPE identities plus structural EOS
Context length	128
Layer count	8
Head count	8
Head width	16
Embedding width	128
Feed-forward multiplier	4
Feed-forward width	512
Normalisation epsilon	1/100
Normalisation enclosure places	4
Activation-grid denominator	256
Trainable exact scalars	1,701,248

The tied language-model head introduces no separate token-projection matrix. Existing proof records retain structural future exclusion and exact serialisation. Architecture size is an engineering choice within the admitted finite family, not a theorem that eight layers or width 128 is universally required. The active build has no paper result until its final receipts and gates exist.

11. Learning constitution

Existing readiness audit v1 states that serving and exact learning observations mutate and invoke the same active native transformer path. This paper adopts that recorded boundary. Causal examples are formed from the reversible conversation serialisation, and the response-only objective retains the user prompt in the forward context while masking it from the assistant-target loss. Every accepted exact update must bind its before and after model identities, development examples, objective, rounding/enclosure records and parent lineage.

A floating process may propose parameters for engineering efficiency. Such a proposal is explicitly non-proof. The current proposal code uses PyTorch floating tensors, conventional square root and softmax, cross-entropy and Adam; these are declared optimisation instruments, not SFT proof scalars or the Discord runtime law. The proposal becomes eligible only after conversion to the exact rational carrier and comparison under the registered exact fixed-grid objective. Exact local coordinate closure retains unchanged, positive-junction and negative-junction candidates. The saved model, tokenizer and receipt form one identified bundle.

That exact acceptance objective is a 40-example causal target-margin check; it does not independently reconstruct the floating teacher soft-mass loss. Useful transfer from the teacher is consequently an empirical question for the frozen conversation and comparator gates, neither of which has a passing active result at this snapshot.

Holdout identities, frozen conversation prompts and manual observations are prohibited from proposal training and exact calibration. A later online learning successor must preserve its parent bundle and complete ordering record. Online learning is

disabled for the first manual observation so that the tested artefact remains fixed.

12. Corpus custody

Two development sources are registered:

1. eight project-authored mechanism dialogues used to exercise exact construction and acceptance; and
2. 10,862 DailyDialog conversations used for broader parameter proposal.

The official DailyDialog train partition contains 11,118 rows. The initial rank rule assigned 2,048 rows to development and ranks 2,048 through 2,303 to a 256-row probe. The current expanded support preserves those same 256 probe identities and admits every other training row, producing 10,862 development rows. The validation and test archives remain unopened.

The development source SHA-256 is

9f8b7e22bbfb9cdfcbaab785a9ba78aef5b3af44fcc2719b742cbfcbd1f9f050; the probe SHA-256 is 12d7c5fb8e7cf66a3538a30ecb8335ae6ad325d6c0dd3a4614f925238245bf26. The outer dataset archive SHA-256 is 2de93416a95060057d8b95ef9e2ab9b17a317906f1e352414001a1887f081193.

DailyDialog remains under the registered CC BY-NC-SA 4.0 boundary: local, non-commercial development only, with attribution and share-alike retained. No row is queried during inference. The model bundle contains learned parameters and a merge grammar, not a response lookup table.

13. GPT-2 teacher distillation and persistence

The official frozen GPT-2 124M model is an offline learning instrument. Its teacher parameter count is 124,439,808 and its weight identity is sha256: 248dfc3911869ec493c76e65bf2fcf7f615828b0254c12b473182f0f81d3a707. The environment is local-only and network-disabled.

Selected development prompts are wrapped as User: <natural prompt>\nAssistant:. At multiple deterministic response prefixes, the complete teacher distribution is contracted into 257 groups: one for each possible first output byte and one for EOS. A deterministic largest-remainder rule places the complete mass on a 16,777,216 = 2²⁴ grid.

The current evidence set records:

Field	Value
Selected assistant-turn examples	8,192
Distinct source dialogue rows	6,121, derived by unique-source_row_id aggregation over the retained signal corpus
Selected soft signals	32,763
Groups per signal	257
Mass-grid denominator	16,777,216
Source assistant targets available	28,050
Prohibited evaluation prompt present	false
Network used	false
Student initialised from teacher	false
Teacher parameters copied	0
Teacher tokenizer serialised	false
Teacher model serialised	false
Native runtime teacher dependency	false

The signal-corpus SHA-256 is b4fb07550506758266ef4f433285d594307cdfcddfa21c336058aefefa19e6bf. The 6,121-row count is not a field in teacher receipt v2. It is independently derived from the retained signal corpus and recorded in audits/continuation_checkpoint.json. The student contracts its own token probabilities by first byte before comparison, so no GPT-2 token identity enters the native model.

Persistence is a protocol requirement. Once accepted, the native bundle must load and infer without GPT-2 or Transformers. A future, larger open teacher may extend the accepted lineage, but only through a separately registered zero-copy receipt. Distillation transfers observed finite behaviour; it does not transfer literal weights or establish proof-equivalence to the teacher.

14. Exact artefact evidence and independent verification

The widest exact artefact is bound by:

- model: sha256: fe8f6f64f8786334d63a0aabdcef260b3fcbcfef0e05f727d07fd39bd4fffbde4d;
- tokenizer: sha256: 09891845445384c92df8fb2cdf300d5e201dce4403781b80f2a8167ef4d3dc90;
- bundle: sha256: 6b460b208438a1b4c71695fa9e700eb5cb2f41b02758946df48bc6ee6a4c81e4.

Its exact learning record contains 40 acceptance examples: eight mechanism examples and 32 external development calibration examples. The 256-row probe was excluded. The registered objective changed from 64676464973/65536 to 5364696355/262144, so the engineering proposal was accepted at that exact boundary.

The separate standard-library verifier loaded the artefact and trace without importing the runtime model. It reconstructed eight layers, eight heads and 1,538 output logits, checked at least 8,192 quantisation records and returned **pass**. Mechanism omission, future-mask leakage, shape tampering, valid-UTF-8 source-sequence round-trip failure, untied-head identity and artefact tampering are adverse conditions. This verifier does not cover tokenizer construction, learning, generation-policy, conversation-state or Discord rows.

This evidence supports the statement: **the identified artefact implements the declared core exact forward path**. It does not support the statement that the artefact is conversationally acceptable.

15. Conversational evidence and retained adverse results

The frozen offline gate uses five single-turn cases and one two-turn exchange, for seven replies in total. Necessary automatic checks require non-empty outputs, exact model identity, terminal reason, latency, direct use of the word “green” in the explicit fact case and use of “green” in the multi-turn follow-up. Cold review additionally rejects malformed language, fragments, irrelevance, contradiction, unusable repetition or truncation.

Eleven native candidates have been retained as adverse hand-off results. Their history includes:

Development stage	Observed boundary	Determination
Minimal exact carriers	Fragmented non-conversation	Rejected
Small scaled carriers	Learned fragments without coherent exchange	Rejected
Broader corpus coverage	Repetition and malformed grammar	Rejected
Corrected speaker grammar	Improved sentence shape but generic fragments	Rejected
Response-only loss	Collapse to high-frequency short replies	Rejected
Mechanism-priority balance	Exact mechanism reproduction without unseen generalisation	Rejected
Eight-layer full-train coverage	Better carrier scale but incoherent replies	Rejected
Extended proposal training	Better sentence shape but weak relevance and no fact grounding	Rejected
Continued training	Plausible reactions but failed direct and multi-turn memory	Rejected
Wider 128-dimensional artefact	Exact mechanism pass, malformed/relevance failures and no green grounding	Rejected

The detailed evidence file preserves all 11 model identities and their recorded outputs and assessments, with terminal reasons and correction records where captured. A controlled sampled-decoding probe on the widest artefact also failed the MVP latency boundary before completing a first reply. No failed candidate is removed because a later candidate may improve.

The current empirical statement is therefore concise: **the core exact forward mechanism is demonstrated for an identified artefact; acceptable unseen conversation is not demonstrated**.

16. Chronology, corrections and historical reconciliation

The chronological boundary is:

1. freeze architecture inventory, SFT dependencies, translation grammar and candidate census;
2. build the tokenizer and native parameters from development sources;
3. freeze teacher-signal custody without evaluation prompts;
4. produce and quantise an engineering proposal;
5. decide the exact objective;
6. run the registered independent verifier;
7. open the development-visible frozen conversation gate;
8. seal a passing candidate before comparator outputs;
9. execute the GPT-2 comparison;
10. rerun final seals and secret checks; and
11. permit Maria's private manual Discord observation.

Several historical corrections remain material:

- V2's split learning path wrote corrections into a structure the serving generator did not use; V3 requires one active model path.
- V2's first visible replies were poor even when delayed content later matched a keyword; V3 evaluates the first visible response.
- An early V3 corpus description named 2,048 proposal rows. The current source-bound receipt preserves the original 256 probe identities and uses all other 10,862 official train rows.
- One early serving route had a speaker-grammar mismatch. Correcting the grammar retained the user's green fact in the prefix but did not repair the learned grounding failure.
- Earlier build receipts could describe quality as not yet measured. The later frozen conversation record lawfully adds an adverse quality determination without changing their exact mechanism result. No post-target observation is backdated into training or a blind result. The active distillation run is excluded from the scientific result until its final receipt exists.

17. Discord application and live terminal

Discord is the MVP transport, not the model. The application admits text only from channel 1523685773998555227. Every other guild channel and every direct message is structurally rejected. Bot messages are ignored, message length is bounded, channel state is ordered, and model processing completes before its successor.

The same channel also functions as the user-visible development terminal. Sanitised structured events may report phases, finite counts, losses, model identities, checkpoints, ETA estimates, acceptance results and controlled failures. Credential-bearing fields and raw bulk teacher masses are prohibited.

At the present boundary, the terminal relay is connected output-only. It posts training progress and automatically refreshes ETA when the estimate is available. It does not accept conversational input because no candidate has passed the offline quality gate. This is active monitoring, not an assertion that the bot is manually testable.

The Discord token is read from the original Unison AI environment only at launch and remains in process memory. It is not copied into this workspace, configuration, logs, evidence or paper. The production adapter has no scripted reply or model-failure fallback.

18. Comparator, promotion and first manual observation

The frozen comparator is the official GPT-2 124M model. The same frozen weights may have supplied teacher signals, but comparator prompts and outputs form a separate sealed partition and cannot re-enter training. The V3 candidate must be sealed before comparator output access.

The comparison retains non-empty completion, malformed output, relevance, contradiction, direct and multi-turn grounding, repetition, termination, latency and memory. Base GPT-2 is a completion model rather than a dialogue-fine-tuned assistant, so disagreement is reported dimension by dimension. It is not automatically an upgrade or regression.

Promotion requires:

Gate	Present status
Final native candidate and bundle	Pending
Registered exact objective	Pending for active candidate
Independent reconstruction	Pending for active candidate
Frozen offline automatic conversation checks	Pending
Cold conversational review	Pending
Frozen GPT-2 comparison or explicit unavailable record	Pending
Final seals, provenance and secret audit	Pending
Private conversational Discord launch	Not run
Maria Smith manual observation	Not run

The manual protocol tests one unregistered greeting, one context-dependent follow-up, silence in another guild channel and silence in direct messages. Possible outcomes are `PASS_MECHANISM`, `FAIL_MECHANISM`, `FAIL_CONVERSATION` and `INCONCLUSIVE`. A mechanism pass does not establish GPT-2-level quality or scientific upgrade.

19. Complete claim and requirement inventory

The application imports 27 direct dependencies and registers 40 operation rows. It does not register a new theorem or law in the frozen SFT admission engine. The appendices retain the complete paper-level claim and evidence inventory.

The principal application requirements are:

Requirement family	Current determination
Fresh V3 boundary; V2 reference-only	Pass
One-to-one operation-role map	Pass at declared finite operation boundary
Operation direct checks	Pass at declared finite operation boundary
Exact native architecture	Pass for identified artefacts
No V2 runtime, copied pretrained parameter or puppetry	Pass for current static and receipt checks
Provenance, census and corpus custody	Pass for current support
Direct, metamorphic, tamper and adverse tests	56 pass; adverse records retained
Independent verification	Parent artefact passes; active candidate pending
Versioned build, corpus, conversation and manual protocols	Pass
Shared identified parameters and registered law	Readiness audit v1 records the same active native transformer path
Single-channel Discord and terminal route	Offline mechanism pass; output-only live relay confirmed
Conversational readiness	Pending; 11 prior candidates rejected
Final seals and secret safety	Shared seals valid; final post-build rerun pending
First manual test	Not run
No public deployment or premature claim	Pass in current state

Requirement family	Current determination
Frozen small-model comparator	Protocol ready; execution pending

20. Reproducibility and verification

Reproduction is layered so that no single success is overstated:

1. verify the frozen SFT engine seal;
2. verify the frozen verification-authority seal;
3. regenerate and compare the 256-form translation census;
4. validate the registered JSON and manifest identities;
5. run the Unison Fold AI workspace tests;
6. inspect the exact build receipt and learning trace;
7. run the registered independent verifier;
8. run the frozen offline conversation gate on the identified candidate;
9. run the comparator only after candidate sealing; and
10. launch Discord only after all offline promotion gates.

The exact commands, prerequisites and expected outcomes are supplied in `paper/REPRODUCIBILITY.md`. Large model artefacts, raw teacher-signal rows, DailyDialog source data, credentials, caches and the live mutable terminal log are excluded from the lightweight publication bundle. Their identities and reconstruction rules remain in the local project evidence.

The latest pre-paper recorded workspace suite contains 56 passing tests and no failures or errors. Those tests establish implementation and protocol behaviour at their declared boundaries. They do not establish conversational validation.

21. Limitations

The present limitations are substantive:

- the unique survivor is unique only within the registered eight-axis translation grammar;
- operation-role correspondence is not numerical equality with GPT-2;
- the verified artefact is one finite configuration, not every configuration;
- rational translations may produce different learned and generated distributions from conventional softmax and GELU;
- the paper does not extend the verification boundary recorded by readiness audit v1;
- the development corpus is English-language and DailyDialog-specific;
- DailyDialog use is restricted to local non-commercial development;
- the frozen offline prompts are development-visible because earlier candidates have already been evaluated on them;
- all 11 observed candidates failed conversational hand-off;
- no final independent verification exists for the active continuation;
- the official GPT-2 comparator has not been executed for a sealed passing candidate;
- no manual Discord conversation has been observed;
- no public deployment, user study, safety study or broader language evaluation has occurred;
- no result establishes consciousness, sentience or phenomenal attention; and
- an application proof cannot admit or modify an SFT law.

These boundaries are not softened by the presence of working Discord wiring, training progress or a favourable loss curve.

22. Open frontier and amendment rule

The active distillation run may continue under the existing registered process. No success is assigned before its final receipt and unchanged gates. A successful continuation must close, in order:

1. final native candidate, bundle and registered exact objective;
2. independent reconstruction of the active candidate;
3. frozen unseen development conversation and cold review;
4. frozen GPT-2 comparator or explicit unavailable classification;
5. final engine, authority, provenance and secret checks;
6. accepted native runtime launch; and
7. Maria Smith's first private manual Discord observation.

Future publication updates will amend this paper with complete identities, outputs, metrics, controls and adverse records. They will preserve version 0.1.0-rc1 as the preliminary chronology. A later larger open teacher may be introduced only after the accepted GPT-2-distilled lineage is persistent and under a separately frozen protocol.

Publication amendment is action-specific. Maria Smith must explicitly authorise each future push, DOI operation, upload or remote publication. The 31 July 2026 instruction authorises this version's GitHub and Zenodo actions only. A request to update the paper does not silently authorise scientific admission.

23. Conclusion

Unison Fold AI has reached an exact but deliberately bounded result. The functional computation of a GPT-2-class causal transformer is represented by 40 registered operation-role translations, supported by 27 direct SFT dependencies. An eight-axis grammar generates 256 complete translation candidates and leaves one unique preserving survivor. An identified eight-layer, eight-head native artefact executes that path, improves the registered exact objective after quantisation and passes independent forward reconstruction.

The major conflation resolved by the programme is the difference between mechanism and conversation. The verified artefact still failed unseen conversation, and it remains rejected for hand-off. Eleven failures are preserved. The response is further learning, not a scripted answer, response table, runtime teacher or copied parameter set.

The active rebuild distils finite behavioural evidence from frozen GPT-2 124M into a fresh native student while copying zero teacher parameters. It has no scientific result in this version and, once accepted, must persist without the teacher. Exact acceptance, independent verification, conversational readiness, comparison and manual Discord evidence remain open for the active candidate.

The current evidence therefore establishes an auditable SFT translation and exact proof-trace implementation mechanism at the boundary recorded by the existing project evidence. No GPT-2 parity, conversational upgrade, newly admitted SFT law or completed conversational model follows. That narrower result is the correct computational-proof statement to carry into the next amendment.

Data and code availability

The human-readable paper, publication-update protocol, guidance-compliance record and reproducibility instructions are maintained under `applications/frontier/v3_computational_proofs/Unison Fold AI/paper/`. Original project code is intended for release under Apache-2.0 after Maria Smith's action-specific authorisation. This paper and original documentation are intended for CC BY 4.0 release.

DailyDialog data remain under CC BY-NC-SA 4.0 and are not relicensed by this paper. GPT-2 weights and source retain their upstream terms. Credentials, local caches, raw bulk teacher distributions, mutable live logs and oversized artefacts are not included in the lightweight publication bundle. Their source and checksum records remain available for authorised local reconstruction.

This version's paper/evidence commit, direct GitHub `main` push and standalone Zenodo publication are authorised. No public deployment of the conversational bot or future remote publication action follows from that authority.

Contribution statement

Maria Smith conceived Smithian Fold Theory, defined the Unison Fold AI programme, set the mathematical and empirical boundaries, directed the development and retains scientific authorship and publication authority. Automated agents assisted with implementation, verification, evidence organisation and editorial preparation. Such assistance carries no scientific authorship, admission authority or publication authority.

References

1. Smith, Maria. *Smithian Fold Theory V3 Constitution, admitted claim packages and machine receipts*. Ernos Labs repository, current local evidence state, 31 July 2026.
2. Smith, Maria. *Unison Fold AI: Mathematical and scientific application boundary; GPT-2-class operation to V3 SFT correspondence*. Ernos Labs, version 0.1.0-rc1 working record, 2026.
3. Radford, Alec; Wu, Jeffrey; Child, Rewon; Luan, David; Amodei, Dario; Sutskever, Ilya. *Language Models are Unsupervised Multitask Learners*. OpenAI, 2019. Source snapshot SHA-256: d9d852e2894556e73f53cb22b7c605a9643d6f0b19bf604b429ed6192fa24f4e.
4. OpenAI. *openai/gpt-2 archived reference implementation*. Frozen revision 9b63575ef42771a015060c964af2c3da4cf7c8ab, accessed through the registered local source record.
5. Li, Yanran; Su, Hui; Shen, Xiaoyu; Li, Wenjie; Cao, Ziqiang; Niu, Shuzi. “DailyDialog: A Manually Labelled Multi-turn Dialogue Dataset”. *Proceedings of IJCNLP 2017*, paper I17 - 1099. Dataset attachment captured under the registered CC BY-NC-SA 4.0 boundary.
6. Discord. *Developer platform and gateway documentation*. Used only for the downstream transport interface; it supplies no model or scientific premise.

Machine-identity appendix

A. Protected SFT authority

Object	Current identity at preparation boundary
Frozen SFT engine Git tree	ad30f4866c18b2adbade95a0b2de40d5caa61308
Frozen SFT engine runtime-byte seal	sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
Frozen verification-authority seal	sha256:bfb10a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
Publication-guidance SHA-256	5ea6b5b558671b0144a553c1b8d20ca32c30b4e01a7d840f6e17248f085405cc

The engine and verification-authority seals passed during publication preparation. They must be rerun after the active candidate is built and before manual launch. This application does not edit either protected surface.

B. Translation and reference identities

Object	Identity
Translation candidate census	sha256:019797d71296fcc4e8dffa878e3c9d84ab8f70eebaa19dd187b28abd89b80346
GPT-2 paper snapshot	sha256:d9d852e2894556e73f53cb22b7c605a9643d6f0b19bf604b429ed6192fa24f4e
GPT-2 repository revision	9b63575ef42771a015060c964af2c3da4cf7c8ab
GPT-2 model.py	sha256:2ff5065f5cac3dc93065f8a3f36eb013e407972a80abcd4ee6b89a7138a0d5a6
GPT-2 encoder.py	sha256:84cadd892377b74de8b61f1fdf2437de9dc025c2f3bc73a981edcb0605e4f9a
GPT-2 sample.py	sha256:7a726916bdc416b296665b52ea7b87581152e16794c8eaff90bf4f89b1235771

C. Data and teacher identities

Object	Identity
DailyDialog outer archive	sha256:2de93416a95060057d8b95ef9e2ab9b17a317906f1e352414001a1887f081193
DailyDialog train archive	sha256:6191257f8250da0e045bfd1d017645e15b2358df975ec4c5c431a5a1f99662a4
Development rows	sha256:9f8b7e22bbfb9cdfcbaab785a9ba78aef5b3af44fcc2719b742cbfcbdd1f9f050
Frozen development probe	sha256:12d7c5fb8e7cf66a3538a30ecb8335ae6ad325d6c0dd3a4614f925238245bf26
GPT-2 124M teacher weights	sha256:248dfc3911869ec493c76e65bf2fcf7f615828b0254c12b473182f0f81d3a707
Teacher signal corpus	sha256:b4fb07550506758266ef4f433285d594307cdfcddfa21c336058aefefa19e6bf

D. Widest exact artefact identities

Object	Identity or exact record
Tokenizer	sha256:09891845445384c92df8fb2cdf300d5e201dce4403781b80f2a8167ef4d3dc90
Model	sha256:fe8f6f64f8786334d63a0aabdcef260b3fcbcfce0e05f727d07fd39bd4ffbbe4d
Bundle	sha256:6b460b208438a1b4c71695fa9e700eb5cb2f41b02758946df48bc6ee6a4c81e4
Objective before	64676464973/65536
Objective after	5364696355/262144
Independent verification	pass; runtime module imported: false
Conversation disposition	accepted_for_handoff: false

E. Publication and active-work identity

Field	Value
Paper version	0.1.0-rc1
Publication classification	Standalone open-access preliminary version; later complete results use Zenodo new-version amendment
DOI	10.5281/zenodo.21726397
Active distilled bundle	Not available at frozen snapshot
Active independent verification	Open
Active conversation decision	Open
Manual Discord observation	Not run
Remote publication	AUTHORISED FOR VERSION 0.1.0-RC1

Final file hashes, PDF identity and Git staging identities are recorded after the paper source is frozen. Embedded source-file hashes in this working draft require refresh at final freeze; this editorial update does not recompute them. No identity is invented for the still-running candidate.